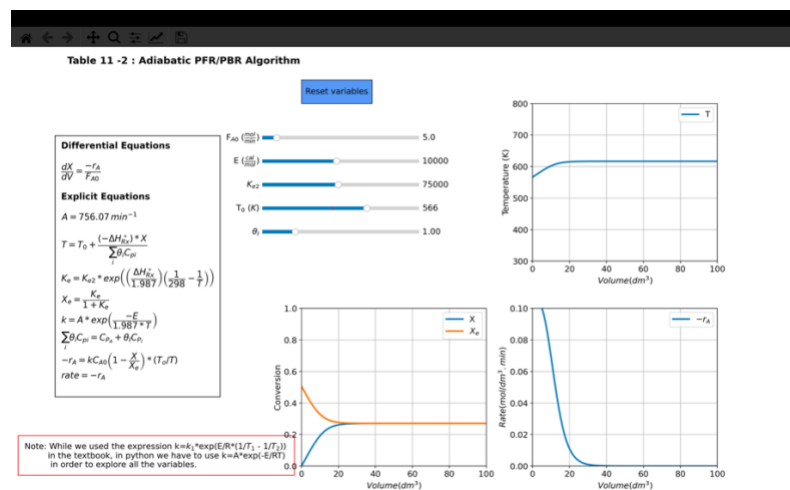
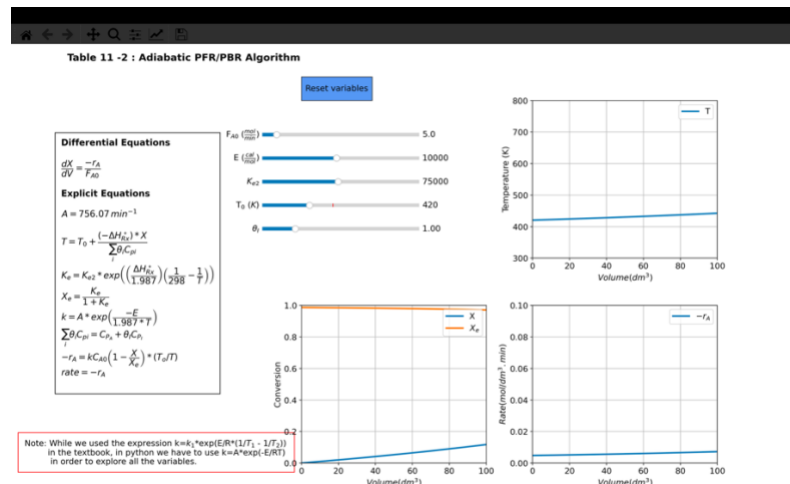
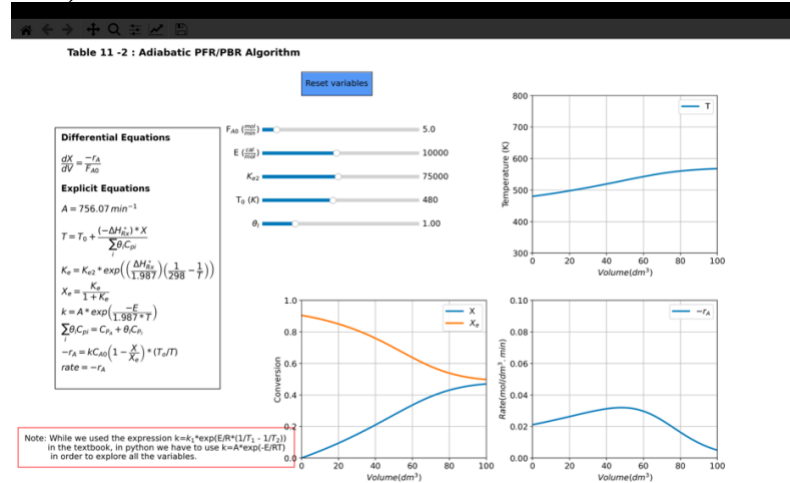


P11-1 a): i)



When you increase the inlet temperature (T0):

1. The equilibrium conversion (X_e) decreases. Because the reaction is exothermic (ΔH is negative), increasing the temperature shifts the equilibrium toward the reactants according to Le Chatelier's Principle.
2. The actual conversion (X) rises more steeply at the beginning of the reactor because the reaction rate constant (k) increases exponentially with temperature, allowing the reaction to proceed faster initially.
3. The profiles meet at a lower conversion value. While the reaction reaches equilibrium faster at higher temperatures, the maximum (X_e) is lower, resulting in a lower final conversion at the reactor exit.

ii)

T_0 has the greatest effect. In an adiabatic system, the temperature at any point is directly tied to the starting temperature. A change in T_0 shifts the entire temperature baseline, whereas K_{e2} and k_1 only change the rate at which the temperature rises toward its adiabatic limit (there's no slider for k_1 , but that is my assumption).

E_a has the greatest effect. Because E_a is located within the exponential term of the Arrhenius equation, even a small numerical change drastically alters the reaction rate constant (k). FA_0 and Θ_{-I} affect the residence time and heat capacity respectively, but their impact is linear or algebraic rather than exponential.

iii)

At low T_0 : The rate starts very low because the molecules lack the kinetic energy to react. The rate may slowly increase as the heat of reaction begins to warm the system, but it remains low overall.

At high T_0 : The initial rate is very high due to the high temperature. However, the rate drops to zero very quickly because the system reaches the lower X_e rapidly.

The peak: Increasing T_0 causes the peak of the reaction rate to shift toward the entrance of the reactor (lower volume) and generally increases the height of that peak.

iv)

1) You have to find the perfect T_0 ; low T_0 means the reaction is too slow, but very high T_0 lowers the final equilibrium conversion. 2) Activation energy has the highest overall impact since a small change in the exponent drastically impacts the rate equation. 3) The reaction rate always spikes as the reactor heats up, but drops down once the conversion gets close to equilibrium.

P11-1 e):

P11-1

e) $\dot{Q} = \dot{n} C_{pw} (T_{out} - T_{in})$

$$\dot{n} = \frac{\dot{Q}}{C_{pw} (T_{out} - T_{in})} = \frac{220000}{18(400 - 270)} = 94.02 \text{ mol/s}$$

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)} = \frac{80 - 60}{\ln(80/60)} = 69.52 \text{ K}$$

$$A = \frac{\dot{Q}}{U \cdot \Delta T_{lm}} = \frac{220000}{100 \cdot 69.52} = 31.64 \text{ m}^2$$

P11-5:

```
import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt

# =====
# Given data
# =====
v0 = 20.0      # dm^3/s
P0 = 10.0      # atm
T0 = 450.0     # K

R = 0.082057   # dm^3 atm / mol / K

# feed is pure A
FA0 = P0 * v0 / (R * T0) # mol/s
thetaA = 1.0
thetaB = 0.0
thetaC = 0.0

# heat capacities (J/mol/K)
CpA = 40.0
CpB = 25.0
CpC = 15.0

# heats of formation at 273 K (J/mol)
HfA = -70.0e3
HfB = -50.0e3
HfC = -40.0e3

# reaction enthalpy
# A -> B + C
dHrxn_273 = (HfB + HfC - HfA) # J/mol

# heat capacity change
DeltaCp = (CpB + CpC - CpA) # J/mol/K

# rate constant parameters
k_ref = 0.133 # dm^3 / (kg-cat s) at 450 K
E = 31.4e3 # J/mol
Rg = 8.314 # J/mol/K

# stoichiometry
# A -> B + C
delta = 1.0 # change in total moles per mole A reacted (1 + 1 - 1 = 1)
eps = delta * 1.0 # pure A feed, so yA0 = 1

def k_of_T(T):
    return k_ref * np.exp((E / Rg) * (1.0 / 450.0 - 1.0 / T))

def deltaH_rxn(T):
    # heats of formation referenced to 273 K
    # DeltaH(T) = DeltaH(273) + DeltaCp*(T - 273)
    return dHrxn_273 + DeltaCp * (T - 273.0)

def concentration_A(X, T):
    # no pressure drop, gas phase, variable density
    # C_A = C_A0 * (1-X)/(1+eps X) * (T0/T)
    CA0 = FA0 / v0
    return CA0 * ((1.0 - X) / (1.0 + eps * X)) * (T0 / T)
```

```

def ODEfun(Y, W):
    X, T = Y

    # numerical protection limits
    X = min(max(X, 0.0), 0.999999)
    T = max(T, 1.0)

    CA = concentration_A(X, T)
    k = k_of_T(T)

    # elementary irreversible rate: -rA' = k CA
    minus_rA_prime = k * CA # mol / (kg-cat s)

    # mole balance
    dXdW = minus_rA_prime / FA0

    # adiabatic energy balance
    FA = FA0 * (1.0 - X)
    FB = FA0 * X
    FC = FA0 * X

    sum_FCp = FA * CpA + FB * CpB + FC * CpC # J/s/K
    dTdW = ((-deltaH_rxn(T)) * minus_rA_prime) / sum_FCp

    return [dXdW, dTdW]

# =====
# Solve ODEs
# =====
Wspan = np.linspace(0.0, 50.0, 500) # kg-cat
Y0 = [0.0, T0]

sol = odeint(ODEfun, Y0, Wspan)

X = sol[:, 0]
T = sol[:, 1]

# derived quantities
CA = concentration_A(X, T)
kvals = k_of_T(T)
minus_rA_prime = kvals * CA

# =====
# Check for 80% conversion
# =====
target = 0.80
idx = np.where(X >= target)[0]

print("-- * 40")
print("REACTOR ANALYSIS RESULTS:")
print("-- * 40")
if len(idx) > 0:
    W_80 = Wspan[idx[0]]
    T_80 = T[idx[0]]
    print(f"80% conversion IS reached at W = {W_80:.2f} kg-cat")
    print(f"Temperature at 80% conversion = {T_80:.2f} K")
else:
    print("80% conversion is NOT reached by W = 50 kg-cat")
    print(f"Maximum conversion reached = {X[-1]:.4f} ({X[-1]*100:.2f}%)")
    print(f"Final temperature at W = 50 kg-cat = {T[-1]:.2f} K")

print("-- * 40")
print(f"FA0 = {FA0:.4f} mol/s")
print(f"CA0 = {FA0/v0:.4f} mol/dm^3")
print(f"DeltaH_rxn at 273 K = {dHrxn_273/1000:.2f} kJ/mol")
print(f"DeltaCp = {DeltaCp:.2f} J/mol/K")
print("-- * 40")

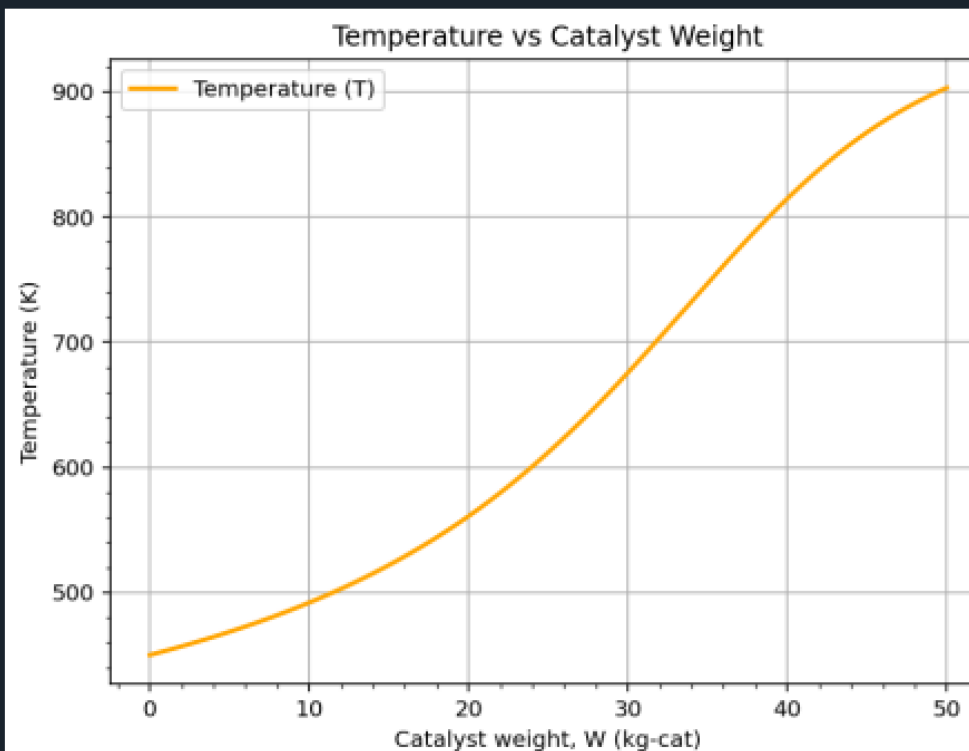
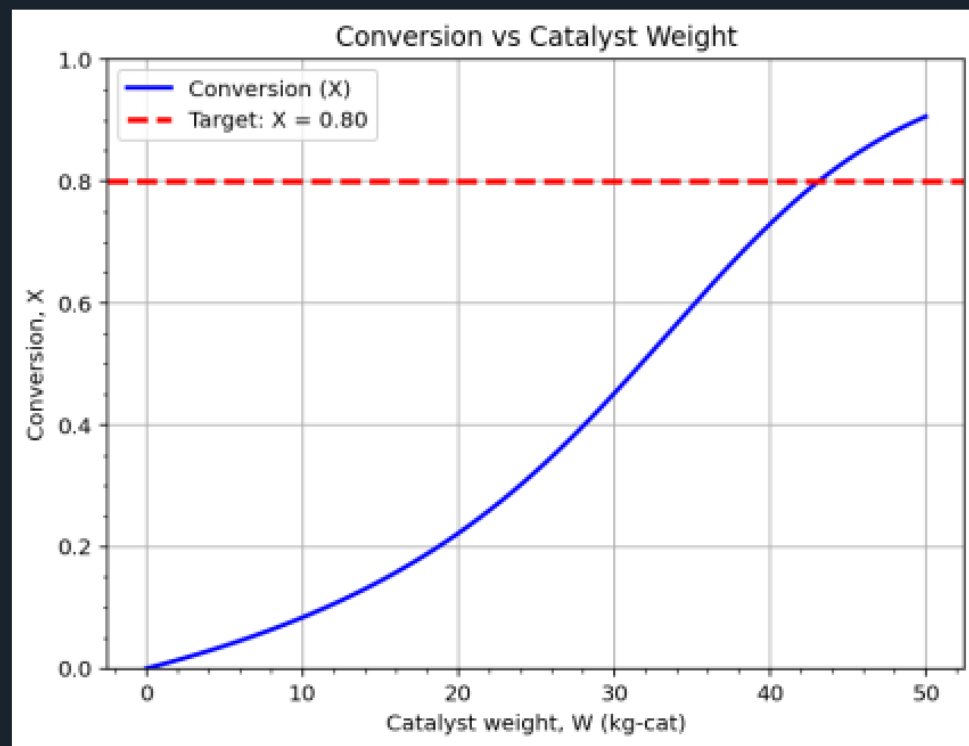
# =====
# Plot X vs W
# =====
fig, ax = plt.subplots(figsize=(7,5))
ax.plot(Wspan, X, label='Conversion (X)', color='blue', linewidth=2)
ax.axhline(0.80, linestyle='--', color='red', label='Target: X = 0.80')
ax.set_xlabel('Catalyst weight, W (kg-cat)')
ax.set_ylabel('Conversion, X')
ax.set_title('Conversion vs Catalyst Weight')
ax.set_ylim(0, 1)
ax.grid(True)
ax.minorticks_on()
ax.legend()
plt.show()

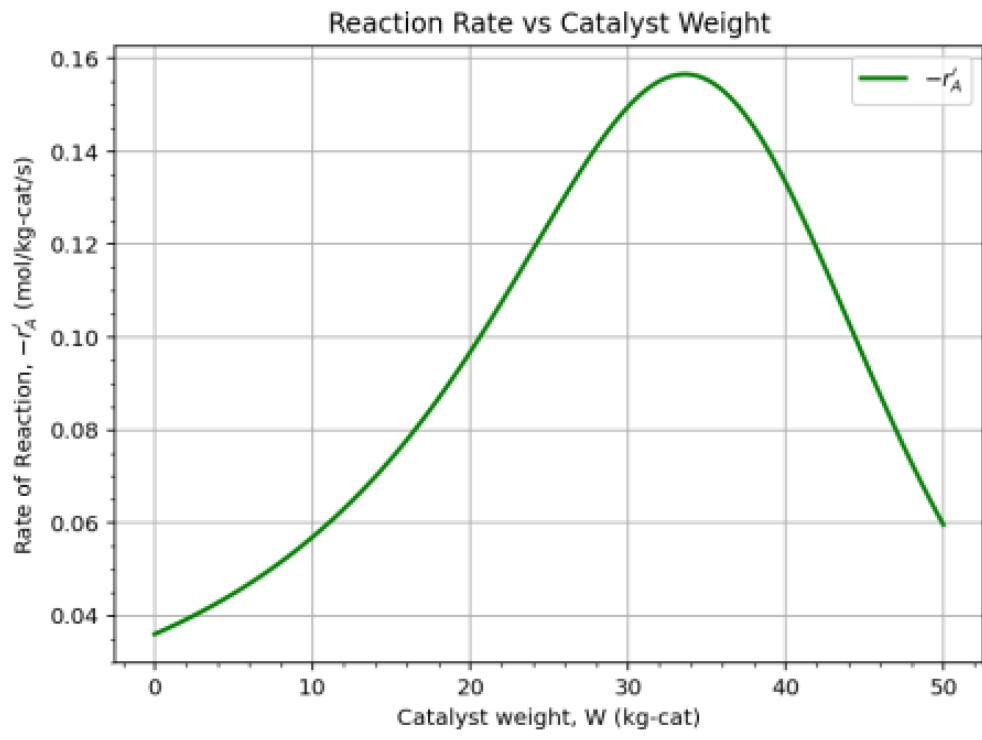
# =====
# Plot T vs W
# =====
fig, ax = plt.subplots(figsize=(7,5))
ax.plot(Wspan, T, label='Temperature (T)', color='orange', linewidth=2)
ax.set_xlabel('Catalyst weight, W (kg-cat)')
ax.set_ylabel('Temperature (K)')
ax.set_title('Temperature vs Catalyst Weight')
ax.grid(True)
ax.minorticks_on()
ax.legend()
plt.show()

# =====
# Plot rate vs W
# =====
fig, ax = plt.subplots(figsize=(7,5))
ax.plot(Wspan, minus_rA_prime, label="$-r_A$", color='green', linewidth=2)
ax.set_xlabel('Catalyst weight, W (kg-cat)')
ax.set_ylabel('Rate of Reaction, $-r_A$ (mol/kg-cat/s)')
ax.set_title('Reaction Rate vs Catalyst Weight')
ax.grid(True)
ax.minorticks_on()
ax.legend()
plt.show()

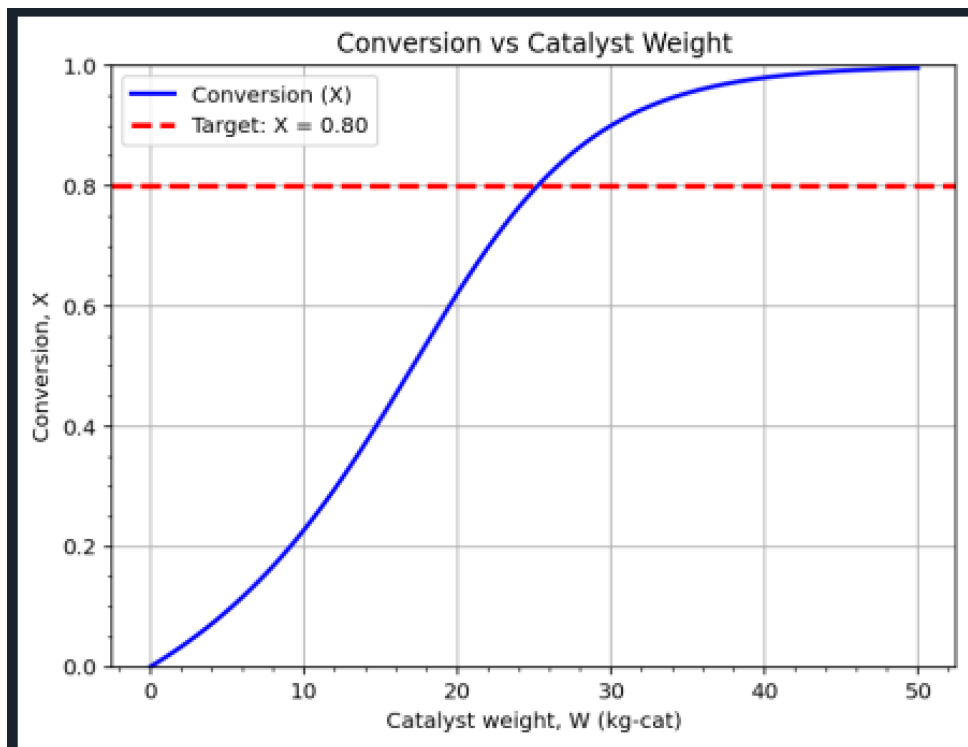
```

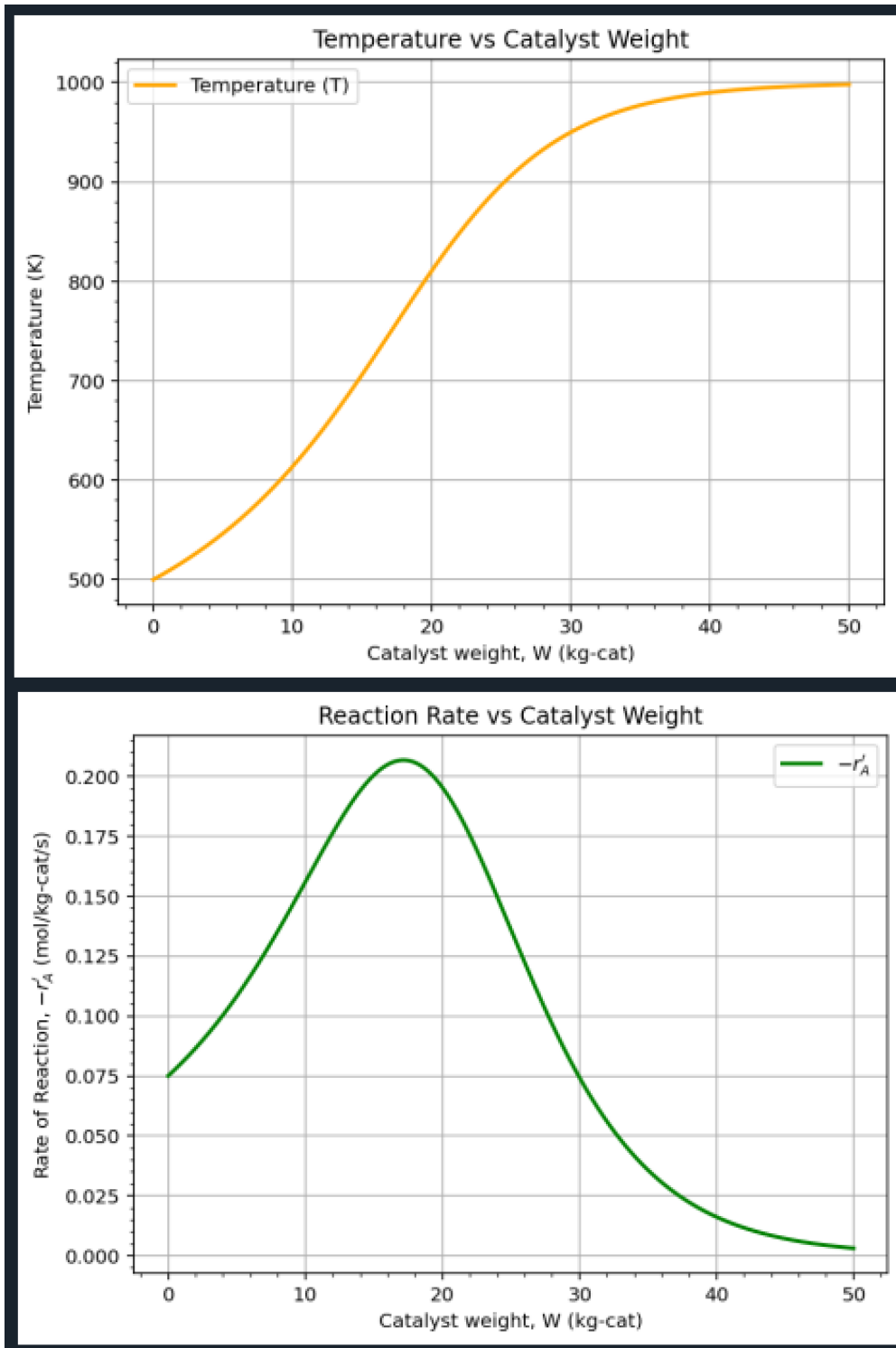
Graphs with the original conditions given in the problem statement:





- a) Because the reactor is adiabatic and the reaction is exothermic, the temperature rises continuously as the reactants are consumed down the length of the PFR. Since this is an irreversible reaction, there is no equilibrium limit, meaning the rising temperature only serves to accelerate the kinetic rate and push the conversion rapidly toward the 80 percent target.
- b) $T=450K \rightarrow T=500K$





Conversion: Accelerates exponentially. You hit the 80% target much earlier, requiring significantly less catalyst. Temperature: Goes up much steeper and reaches its maximum earlier in the reactor. Reaction Rate: Spikes much higher and sooner due to faster initial kinetics, then goes to zero quickly because the reactants are consumed so fast.

c)

```
import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt

# Given data
v0 = 20.0          # dm^3/s
P0 = 10.0          # atm
T0 = 450.0         # K (Isothermal constant temperature)
R = 0.082057       # dm^3 atm / mol / K

# Feed
FA0 = P0 * v0 / (R * T0)  # mol/s

# Heats of formation (J/mol)
HfA = -70.0e3
HfB = -50.0e3
HfC = -40.0e3
dHrxn = (HfB + HfC - HfA)  # -20000 J/mol

# Kinetics
k_ref = 0.133       # dm^3 / (kg-cat s) at 450 K
eps = 1.0           # change in total moles per mole A reacted (pure A feed)
CA0 = FA0 / v0

# ODE system for Isothermal PFR
def ODEfun(X, W):
    X = min(max(X, 0.0), 0.9999)

    # Isothermal concentration (T/T0 = 1)
    CA = CA0 * ((1.0 - X) / (1.0 + eps * X))

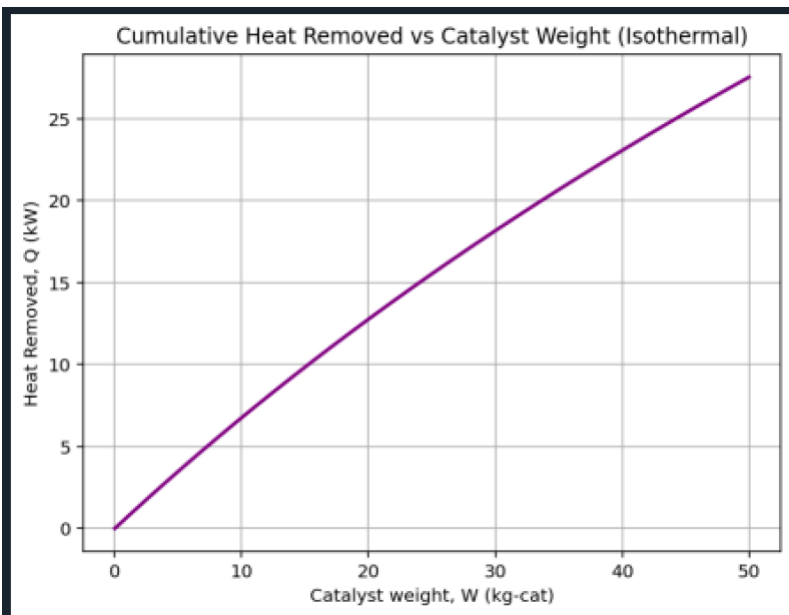
    # Rate is only dependent on concentration now since T is constant
    minus_rA_prime = k_ref * CA

    dXdW = minus_rA_prime / FA0
    return dXdW

# Solve ODE
Wspan = np.linspace(0.0, 50.0, 500)
X0 = 0.0
sol = odeint(ODEfun, X0, Wspan)
X = sol[:, 0]

# Calculate heat removed (kW)
# Q_removed = FA0 * X * (-dHrxn) / 1000 (to convert J/s to kW)
Q_removed = FA0 * X * (-dHrxn) / 1000.0

# Plotting
fig, ax = plt.subplots(figsize=(7,5))
ax.plot(Wspan, Q_removed, color='purple', linewidth=2)
ax.set_xlabel('Catalyst weight, W (kg-cat)')
ax.set_ylabel('Heat Removed, Q (kW)')
ax.set_title('Cumulative Heat Removed vs Catalyst Weight (Isothermal)')
ax.grid(True)
```



For an isothermal reactor, the temperature does not rise, meaning the reaction rate constant (k) does not increase exponentially like it did in the adiabatic case. The reaction rate simply slows down as the reactant A is consumed.

d) $\alpha=0.019$

```
import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt

# =====
# Given data
# =====
v0 = 20.0          # dm^3/s
P0 = 10.0          # atm
T0 = 450.0         # K
R = 0.082057       # dm^3 atm / mol / K

# feed is pure A
FA0 = P0 * v0 / (R * T0)  # mol/s
CA0 = FA0 / v0

# stoichiometry
eps = 1.0          # change in total moles per mole A reacted (pure A feed)

# Pressure drop parameter (Particle size D1)
alpha = 0.019      # 1 / kg-cat

# =====
# ODE system: Y = [X, y]
# independent variable = catalyst weight W
# =====
def ODEfun(Y, W):
    X, y = Y

    # numerical protection
    X = min(max(X, 0.0), 0.999999)
    # Prevent y from dropping below zero (reactor choke)
    y = max(y, 1e-4)

    # Algebraic energy balance (valid because Delta Cp = 0)
    T = T0 + 500.0 * X

    # Rate constant
    k = 0.133 * np.exp((31400.0 / 8.314) * (1.0 / 450.0 - 1.0 / T))

    # Concentration with pressure drop factor (y)
    CA = CA0 * ((1.0 - X) / (1.0 + eps * X)) * y * (T0 / T)

    # elementary irreversible rate: -rA' = k CA
    minus_rA_prime = k * CA

    # Mole balance
    dXdW = minus_rA_prime / FA0

    # Momentum balance (Pressure drop)
    dydW = -(alpha / (2.0 * y)) * (1.0 + eps * X) * (T / T0)

    return [dXdW, dydW]
```

```

# =====
# Solve ODEs
# =====
Wspan = np.linspace(0.0, 50.0, 500) # kg-cat
Y0 = [0.0, 1.0] # initial X = 0, initial y = P/P0 = 1.0

sol = odeint(ODEfun, Y0, Wspan)

X = sol[:, 0]
y = sol[:, 1]
T = T0 + 500.0 * X

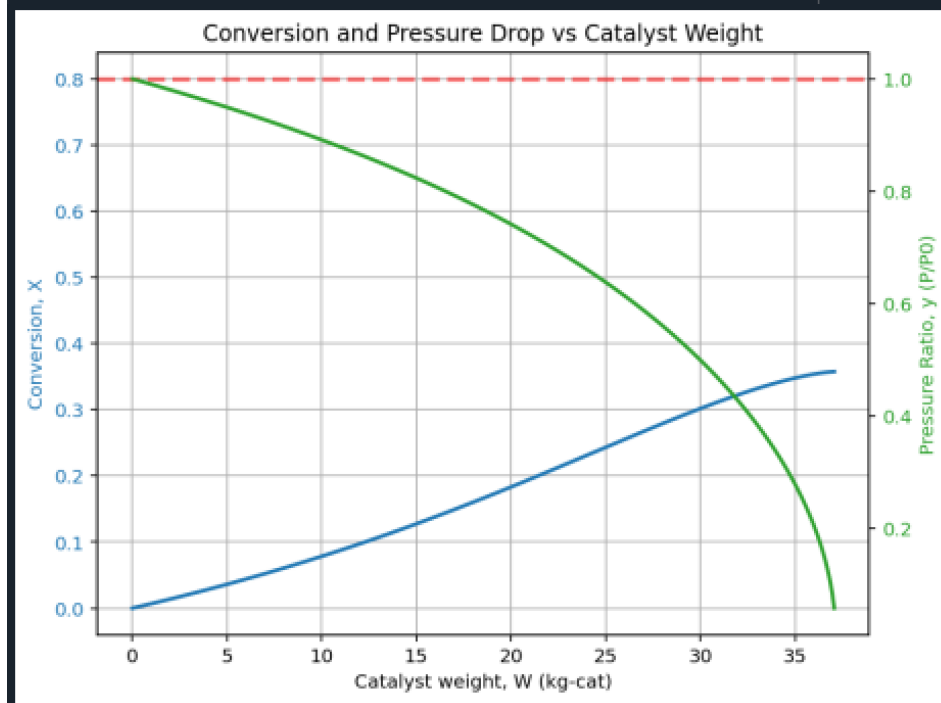
# Check if reactor choked (pressure dropped to 0)
choked_idx = np.where(y <= 0.01)[0]
if len(choked_idx) > 0:
    cutoff = choked_idx[0]
    print(f"WARNING: Reactor choked (Pressure hit 0) at W = {Wspan[cutoff]:.2f} kg")
    # Trim arrays for plotting to stop at the choke point
    Wspan = Wspan[:cutoff]
    X = X[:cutoff]
    y = y[:cutoff]
    T = T[:cutoff]

# =====
# Check for 80% conversion
# =====
target = 0.80
idx = np.where(X >= target)[0]

print("-" * 40)
print(f"RESULTS FOR ALPHA = {alpha}")
print("-" * 40)
if len(idx) > 0:
    print(f"80% conversion reached at W = {Wspan[idx[0]]:.2f} kg-cat")
else:
    print("80% conversion is NOT reached.")
    print(f"Maximum conversion reached = {X[-1]:.4f}")
    print(f"Final pressure ratio (P/P0) = {y[-1]:.4f}")

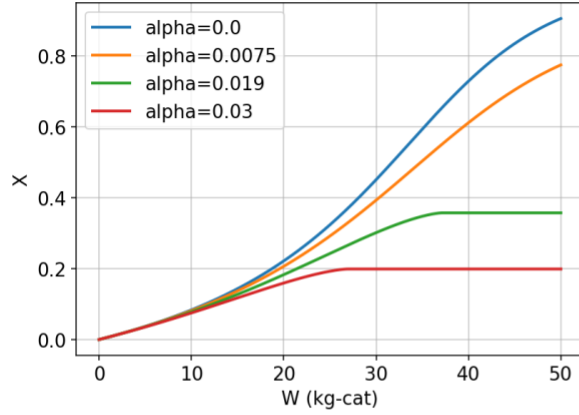
# =====
# Plot X and y vs W
# =====
fig, ax1 = plt.subplots(figsize=(7,5))

```

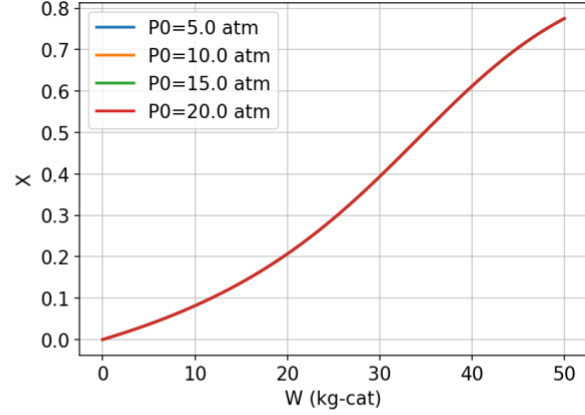


e)

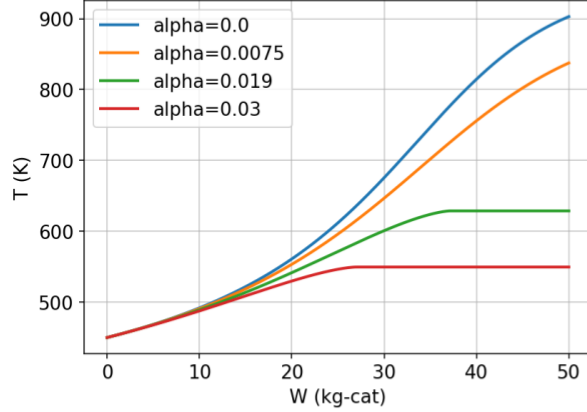
Part 5: Effect of alpha on Conversion



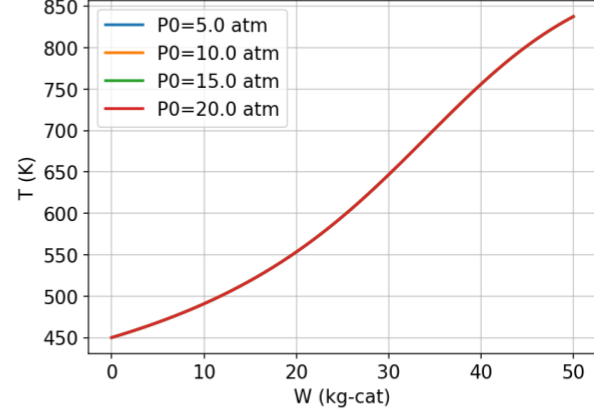
Part 5: Effect of P0 on Conversion (alpha=0.0075)



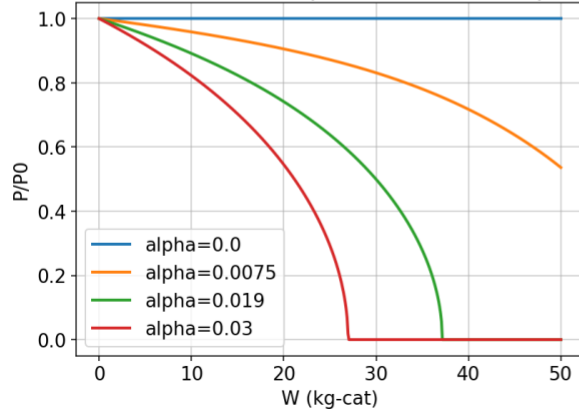
Part 5: Effect of alpha on Temperature



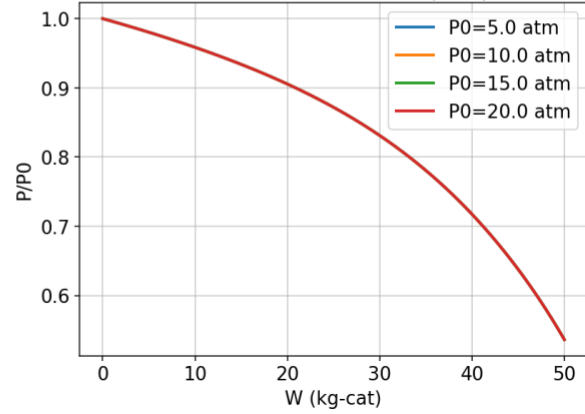
Part 5: Effect of P0 on Temperature (alpha=0.0075)



Part 5: Effect of alpha on Pressure Drop



Part 5: Effect of P0 on Pressure Drop (alpha=0.0075)



P11-7)
a)

P11-7)

$$\frac{dX}{dV} = -r_A / F_{A0}$$

$$-r_A = k (C_A - C_B / K_c)$$

$$k = k_1 (T_0) \cdot \exp \left[\left(\frac{E}{R} \right) \cdot \left(\frac{1}{T_0} - \frac{1}{T} \right) \right]$$

$$C_A = C_{A0} (1-X) \cdot p \cdot (T_0/T)$$

$$C_B = C_{A0} \cdot X \cdot p \cdot (T_0/T)$$

$$-r_A = k C_{A0} \cdot p \cdot (T_0/T) \cdot (1-X - X/K_c)$$

$$X_e = \frac{K_c}{1+K_c}$$

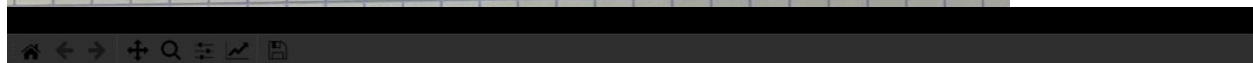
$$\sum C_p = C_{PA} + \theta_1 \cdot C_{p1}$$

$$\sum C_p = 150 + 2 \cdot 18 = 196 \text{ cal/mol/K}$$

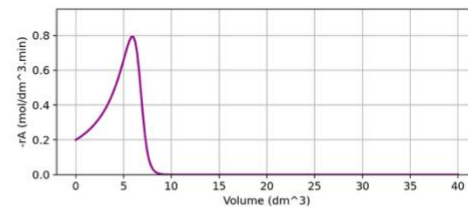
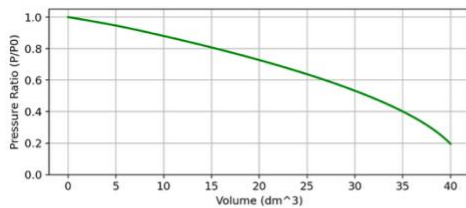
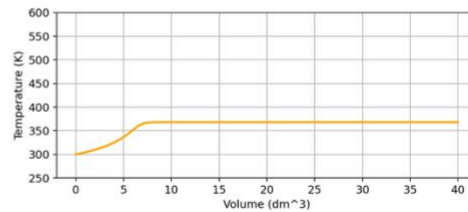
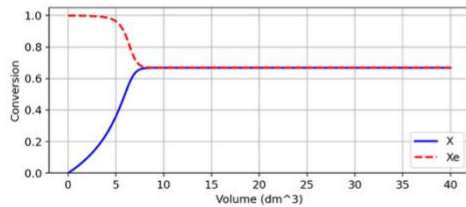
$$T = T_0 + (-\Delta H R_x / \sum C_p) \cdot X$$

$$T = 300 + (20000/196) \cdot X$$

$$\frac{dp}{dV} = - \left(\frac{dp_b}{2p} \right) \left(\frac{T}{T_0} \right)$$



Adiabatic Reversible PBR Simulator



T_0 (K) 300
 θ_1 2.0
 F_{A0} 5.0
 αp_b 0.020

Reset

Temperature (T): The curve rises steadily. This is an exothermic reaction, meaning it releases heat as it converts A to B. Because the reactor is adiabatic, there is no cooling system to remove this heat, so the temperature of the gas naturally goes up as the reaction progresses down the length of the reactor.

Equilibrium Conversion (X_e): The curve drops continuously. According to Le Chatelier's principle, adding heat to an exothermic reaction shifts the equilibrium backward toward the reactants.

Actual Conversion (X): The curve rises quickly at first and then flatlines. The rising temperature initially helps the reaction go faster by increasing the kinetic rate constant. Once X reaches X_e , the system hits equilibrium and the reaction stops, causing the conversion curve to flatline.

Reaction Rate ($-r_A$): The curve eventually goes to zero. The rate depends on the driving force, which is the gap between the actual conversion and the equilibrium limit. As X gets closer to X_e , this gap disappears. When the curves intersect, the forward and reverse reaction rates are equal, making the net reaction rate zero.

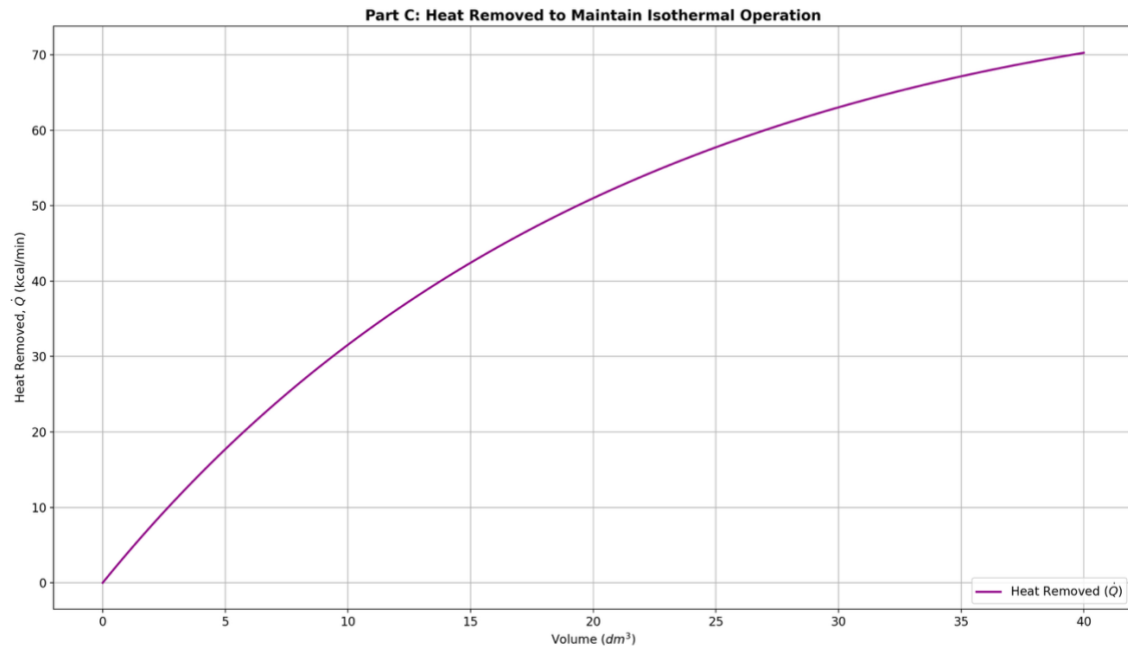
Pressure Ratio (P/P_0): The curve drops steadily downwards. As the gas flows through the solid catalyst particles packed inside the tube, it loses energy, causing the pressure to steadily decrease down the volume of the reactor.

b)

Varying the ratio of inerts (Θ_I): When you increase the ratio of inerts, the temperature curve becomes much flatter and reaches a lower maximum value. Because the temperature does not rise as much X_e curve does not drop as steeply and stays higher down the length of the reactor. As a result, X curve intersects the X_e curve much later, resulting in a significantly higher final conversion.

Varying the entering temperature (T_0): When you increase the entering temperature, the entire temperature curve shifts upward. This causes X_e curve to shift downward, starting at a lower value. X curve initially rises much more steeply, but because the X_e curve is lowered, the two curves intersect very early in the reactor volume. This causes the X curve to flatline much sooner and at a lower final conversion.

c) $Q_{\text{removed}} = F_{A0} * X * (-\Delta H_{rx})$



```
import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt
import matplotlib as mpl

# Set plot parameters for readability
mpl.rcParams.update({'font.size': 12, 'lines.linewidth': 2.0})

# =====
# Given Parameters
# =====
FA0 = 5.0          # mol/min
CA0 = 2.0          # mol/dm³
dH_Rx = -20000.0   # cal/mol
k = 0.1            # min⁻¹ (Constant at 300K)
KC = 1000.0        # Constant at 300K
alpha_rhob = 0.02  # dm³~3

# =====
# Differential Equations (Isothermal)
# =====
def ODEfun(Y, V):
    X, y = Y

    y = max(y, 1e-4)
    X = min(max(X, 0.0), 0.9999)

    # Isothermal Rate Law (T/T0 = 1)
    ra = -k * CA0 * y * (1.0 - X - X / KC)
    minus_ra = -ra

    # Mole Balance
    dXdV = minus_ra / FA0

    dydV = -(alpha_rhob / (2.0 * y))

    return [dXdV, dydV]

# =====
# Solve the System
# =====
Vspan = np.linspace(0, 40, 200) # Volume from 0 to 40 dm³
Y0 = [0.0, 1.0]                # Initial X = 0, y = 1.0

sol = odeint(ODEfun, Y0, Vspan)
X_out = sol[:, 0]

# Calculate Cumulative Heat Removed (in kcal/min)
# Q_removed = FA0 * X * (-dH_Rx) / 1000
Q_removed = FA0 * X_out * (-dH_Rx) / 1000.0
```

P11-9:

P11-9:

$$K(T) = \frac{C_C C_D}{C_A C_B} = K(T_{ref}) \exp \left[\frac{-\Delta_r H}{R} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \right]$$

$$C_A = C_{A0} - C_{A0}X = C_{A0}(1-X)$$

$$C_B = C_{A0}(1-X)$$

$$C_C = C_{A0}X$$

$$C_D = C_{A0}X$$

$$K(T) = \frac{(C_{A0}X)^2}{(C_{A0}(1-X))^2} = \frac{X^2}{(1-X)^2}$$

$$X_e = \frac{\sqrt{K}}{1+\sqrt{K}}$$

$$F_A = F_{A0} - F_{A0}X$$

$$F_B = F_{B0} - F_{A0}X = F_{A0} - F_{A0}X$$

$$F_C = F_{A0}X$$

$$F_D = F_{A0}X$$

$$0 = \sum_{\alpha} F_{\alpha 0} h_{\alpha}(T_0) - \sum_{\alpha} (F_{\alpha 0} + \nu_{\alpha} F_{A0}X)(h_{\alpha}(T))$$

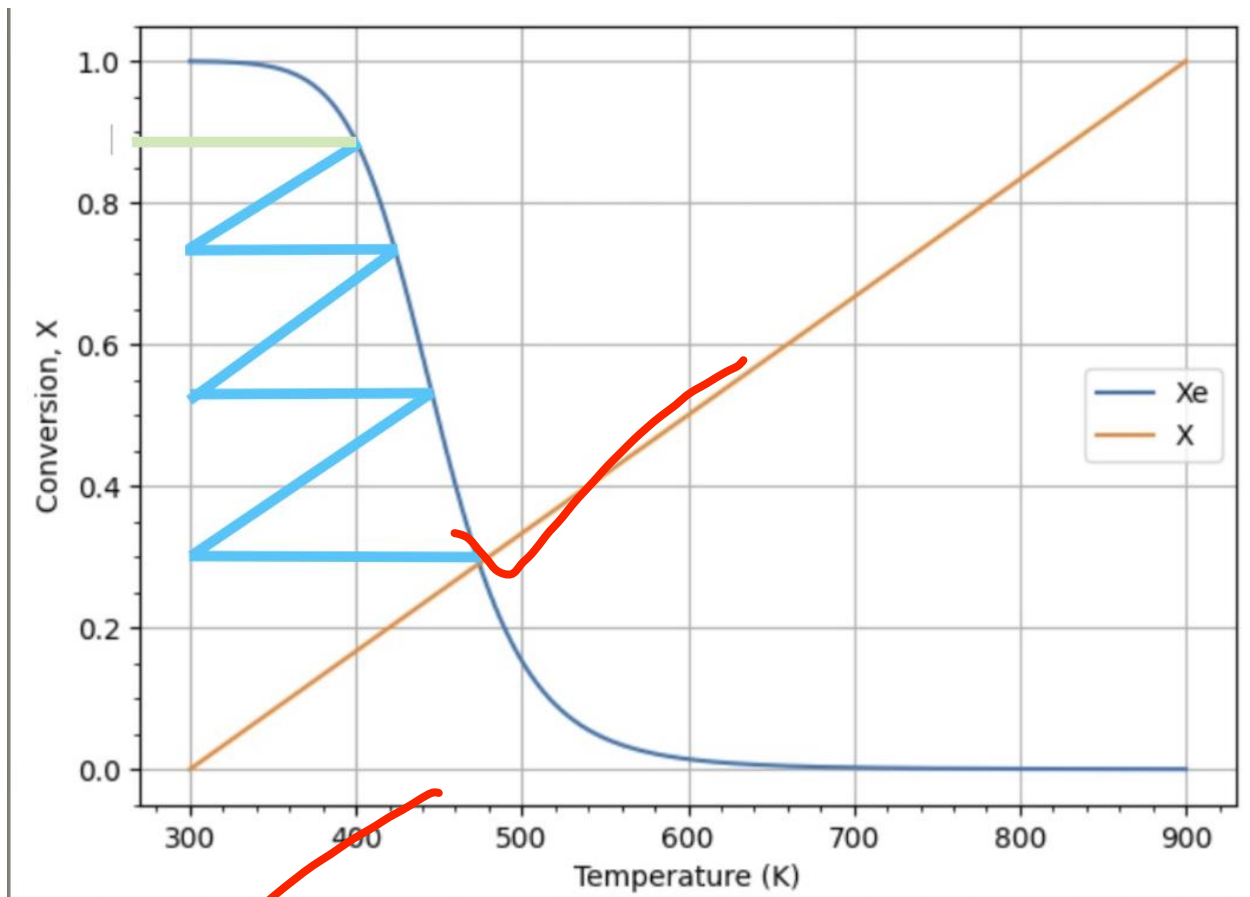
$$0 = \sum_{\alpha} F_{\alpha 0} (h_{\alpha}(T_0) - h_{\alpha}(T)) - \sum_{\alpha} \nu_{\alpha} h_{\alpha}(T) F_{A0}X$$

$$0 = \left(\sum_{\alpha} F_{\alpha 0} - C_{P\alpha} \right) (T_0 - T) - \Delta_r H F_{A0}X$$

$$T = T_0 - \left(\frac{\Delta_r H F_{A0}}{\sum_{\alpha} F_{\alpha 0} C_{P\alpha}} \right) X$$

$$T = 300 + \frac{(30000 \cdot 10)}{500} X = 300 + 600X$$

$$X = \frac{T - 300}{600}$$



Max X is around 0.9.